

King County Housing Market Analysis

Client: Nicole Johnson

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Objective:



Provide data-driven recommendations to help Nicole purchase a house in King County within a moderate budget.



Washington State
map courtesy US Census Bureau

Business Problem & Client

Client profile

- Looking for a central neighborhood
- Moderate budget
- Planning to buy within the next year



Research Questions

1. Does living area affect price?
2. Does location affect price?
3. Does purchase timing affect price?



Dataset Overview

Dataset Summary

~21,000 housing transactions
King County, Washington
Property characteristics + sales data
Target variable: Price

Data Cleaning

Missing waterfront values → filled with 0
Missing basement values → filled with 0
Missing renovation years → filled with 0
Missing view ratings → filled with 0

```
Data columns (total 21 columns):  
#   Column          Non-Null Count  Dtype  
---  -  
5   floors           21597 non-null    float64  
6   waterfront       19206 non-null    float64  
7   view             21534 non-null    float64  
13  yr_renovated     17749 non-null    float64
```

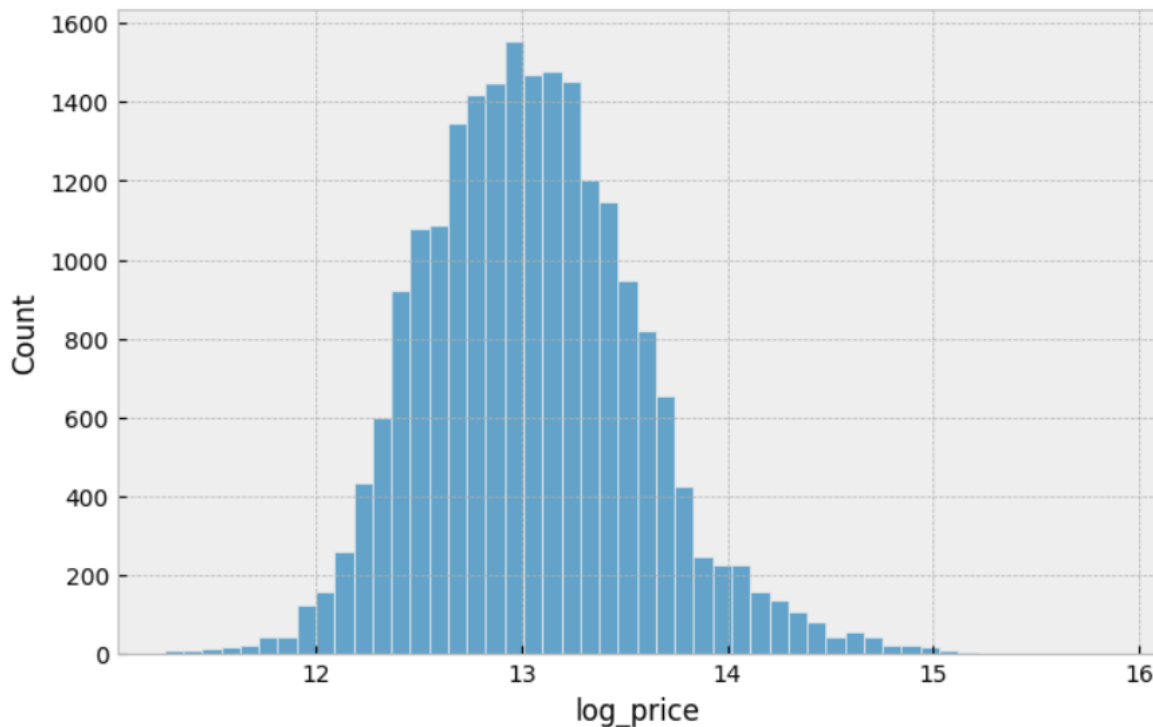


Price Distribution

Housing Market Distribution

Key Findings

- Original distribution highly skewed
- Luxury properties distort market view
- Log transformation reveals typical market behavior



Hypothesis 1

Does Living Area Affect Price?

Findings

- Strong positive relationship
- Larger houses generally cost more
- Living area is a major price driver

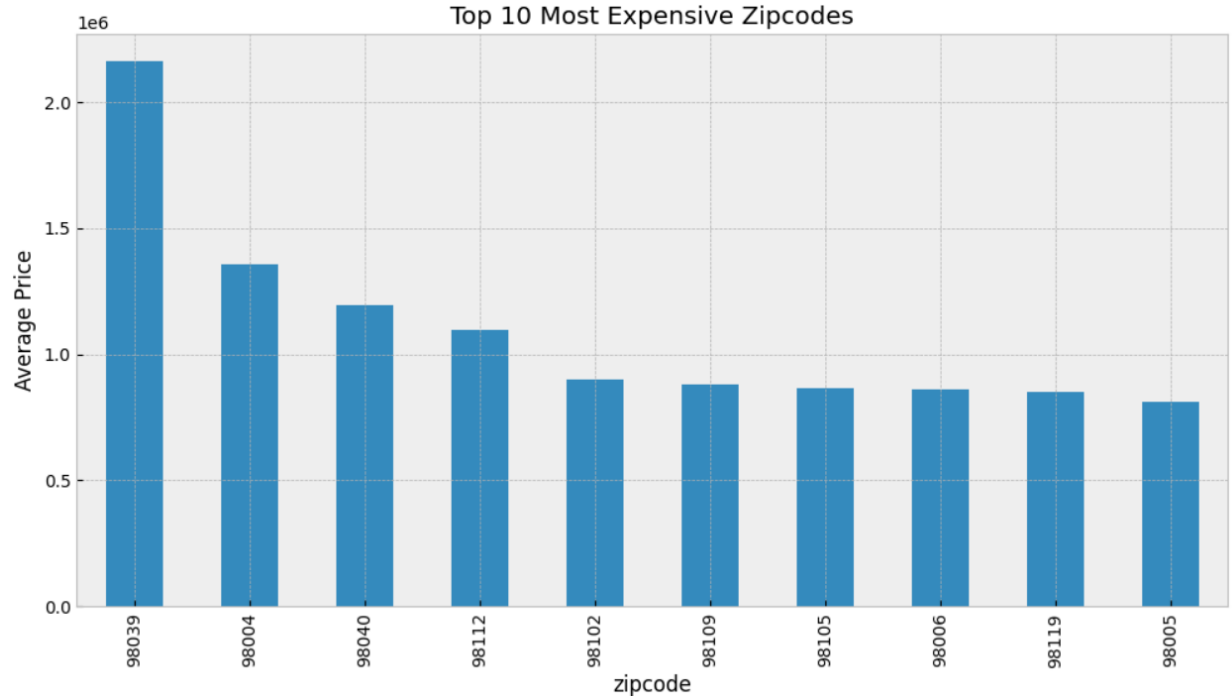


Hypothesis 2 (Geographical Analysis)

Does Location Affect Price?

Findings

- Significant variation across zipcodes
- Some areas command much higher prices
- Location strongly influences affordability

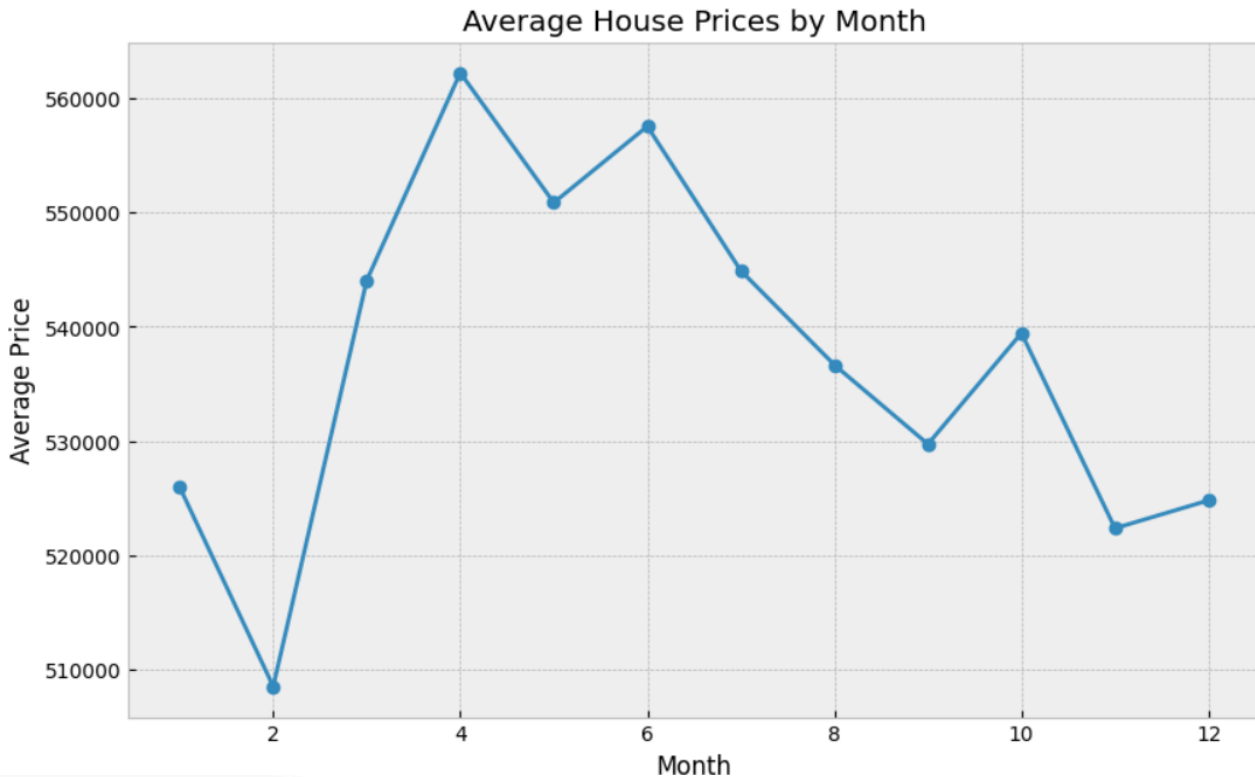


Hypothesis 3

Does Purchase Timing Affect House Prices?

Key Findings

- Average prices vary throughout the year.
- The highest average prices occur around April and June.
- The lowest average prices occur around February, with another decline near November.
- The difference between peak and low months is approximately \$50,000.



Recommendation for Nicole Johnson

- Target moderately priced central-adjacent neighborhoods to balance affordability and accessibility.
- Prioritize living area over bedroom count, as living space has a stronger impact on property value.
- Consider purchasing during lower-priced months (e.g., February or November) when market prices tend to be lower.
- Focus on well-maintained or renovated homes to maximize long-term value and reduce future renovation costs

